Eric D. Sun

Department of Biological Engineering, Massachusetts Institute of Technology
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Education

Stanford University 2020-2025
PhD, Biomedical Informatics

Harvard University 2019-2020
SM, Applied Mathematics

Harvard University 2016-2020
AB, Chemistry and Physics, *Summa Cum Laude*

Professional Appointments

Massachusetts Institute of Technology, Cambridge, MA
- Assistant Professor, Department of Biological Engineering 2026-Present
- Member, The Ragon Institute of MGB, MIT, and Harvard 2026-Present

Stanford University (Zou & Brunet Labs), Stanford, CA
- Postdoctoral fellow, Department of Genetics 2025
- Graduate research assistant, Department of Biomedical Data Science 2020-2025

Harvard University (Mahadevan Lab), Cambridge, MA
- Undergraduate research assistant, Department of Physics and Applied Mathematics 2018-2020

Stanford University (Snyder Lab), Stanford, CA
- Summer research intern, Department of Genetics 2019

National Institute on Aging, Baltimore, MD
- Summer research intern, Human Genetics Unit 2018-2019

Radcliffe Institute for Advanced Study, Cambridge, MA
- Research partner, Zurzolo group 2018-2019
- Research partner, Lenormand group 2017-2018

Harvard Medical School (Sinclair Lab), Boston, MA
- Undergraduate research assistant, Department of Genetics 2017-2018

Icahn School of Medicine at Mount Sinai (Karr Lab), New York, NY
- Winter research intern, Department of Genetics 2017

Weizmann Institute of Science, Rehovot, Israel
- Summer Science Student, Department of Brain Sciences 2016

National Institute on Aging, Baltimore, MD
- Summer research intern, Image Informatics and Computational Biology Unit 2014

Honors and Awards

Forbeck Scholar	2026
Scialog Fellow (Neurobiology and Changing Ecosystems)	2026
International Birnstiel Award for Doctoral Research in Molecular Life Sciences	2025
Rising Stars in Data Science	2025
Rising Star - Genomic Press	2025
National Science Foundation (NSF) Graduate Research Fellowship	2022-2025
Knight-Hennessy Scholar	2020-2025
Bay Area Aging Meeting - Most Outstanding Poster Prize	2024
Paul & Daisy Soros Fellowship for New Americans	2020-2022
Sophia Freund Prize (highest rank graduate in Harvard Class of 2020)	2020
John Harvard Scholar (four-time)	2017-2020
Phi Beta Kappa (Harvard Senior 48)	2019
Radcliffe Research Poster Contest Winner	2018
Amgen Scholar	2018
Detur Book Prize	2017
National Institute on Aging (NIA) Barbara A. Hughes Award of Excellence	2014

Publications

* denotes equal contribution

Select Primary Papers:

1. **Sun ED***, Buendia A*, Brunet A, Zou J. SpatialProp: tissue perturbation modeling with spatially resolved single-cell transcriptomics. Under review.
2. **Sun ED**, Zhou OY, Hauptschein M, Rappoport N, Xu L, Navarro Negredo P, Ling L, Rando TA, Zou J*, Brunet A*. Spatial transcriptomic clocks reveal cell proximity effects in brain ageing. **Nature** (2025).
3. **Sun ED**, Ma R, Zou J. SPRITE: improving spatial gene expression imputation with gene and cell networks. **Bioinformatics** 40, i521–i528 (2024).
4. **Sun ED**, Ma R, Navarro Negredo P, Brunet A, Zou J. TISSUE: uncertainty-calibrated prediction of single-cell spatial transcriptomics improves downstream analyses. **Nature Methods** 21, 444–454 (2024).
5. **Sun ED**, Ma R, Zou J. Dynamic visualization of high-dimensional data. **Nature Computational Science** 3, 86–100 (2023).
6. Buckley MT*, **Sun ED***, George BM, Liu L, Schaum N, Xu L, Reyes JM, Goodell MA, Weissman IL, Wyss-Coray T, Rando TA, Brunet A. Cell type-specific aging clocks to quantify aging and rejuvenation in regenerative regions of the brain. **Nature Aging** 3, 121–137 (2023).
7. **Sun ED**, Qian Y, Oppong R, Butler TJ, Zhao J, Chen BH, Tanaka T, Kang J, Sidore C, Cucca F, Bandinelli S, Abecasis GR, Gorospe M, Ferrucci L, Schlessinger D, Goldberg I, Ding J. Predicting physiological aging rates from a range of quantitative traits using machine learning. **Aging** 13, 23471–23516 (2021).
8. **Sun ED**, Dekel R. ImageNet-trained deep neural networks exhibit illusion-like response to the Scintillating grid. **Journal of Vision** 21, 15 (2021).
9. **Sun ED***, Michaels TCT*, Mahadevan L. Optimal control of aging in complex networks. **Proceedings of the National Academy of Sciences** 117, 20404–20410 (2020).

Other Primary Papers:

10. Liu T, Hao M, **Sun ED**, Markov Y, Zhang L, Zou J, Zhao H. BrainBridge characterizes key factors affecting Alzheimer's disease and associated phenotypes. Under review

11. Su Y, Zhou Y, Wang L, Gullapalli R, **Sun ED**, Sun Y, Zhang M, Hu P, Nauen D, Brunet A, Ming G, Song H. Transcriptomic age clocks of glial cell types in the human hippocampus across the postnatal lifespan. Under review

12. Miklas JW, Papsdorf K, **Sun ED**, Haseley NR, Medoh UN, Tsenter A, McCarthy F, Artiles KL, Hims A, Levina A, Zhou OY, Elias JE, Abu-Remaileh M, Brunet A. Quantitative characterization of the aging intestine secretome reveals novel secreted proteins that act in the extracellular space to regulate lifespan. In revision

13. Alber S*, Chen B*, **Sun ED**, Markov Y, Isakova A, Wilk AJ, Zou J. CellVoyager: AI CompBio agent generates new insights by autonomously analyzing biological data. Accepted at **Nature Methods**

14. Negredo PN, Hauptschein M, Richard D, Abhiraman G, Ramirez-Matias J, Zhou O, Malacon K, Buckley M, **Sun ED**, Xu L, Picton L, Saxton R, Fernandes R, Garcia K, Brunet A. Targeting immune cells in the aged brain with engineered proteins. Accepted at **Immunity**

15. Bianchi F*, Queen O*, Thakkar N, **Sun ED**, Zou J. Exploring the use of AI authors and reviewers at Agents4Science. **Nature Biotechnology** 44, 11–14 (2026)

16. Ruetz TJ, Pogson AN*, Kashiwagi CM*, Gagnon SD, Morton B, **Sun ED**, Na J, Yeo RW, Leeman DS, Morgens DW, Tsui CK, Li A, Bassik MC, Brunet A. CRISPR–Cas9 screens reveal regulators of ageing in neural stem cells. **Nature** 634, 1150–1159 (2024).

17. Xu L, Ramirez-Matias J, Hauptschein M, **Sun ED**, Lunger JC, Buckley MT, Brunet A. Restoration of neuronal progenitors by partial reprogramming in the aged neurogenic niche. **Nature Aging** 4, 546–567 (2024).

18. Ma R, **Sun ED**, Donoho D, Zou J. Is your data alignable? Principled and interpretable alignability testing and integration of single-cell data. **Proceedings of the National Academy of Sciences** 121, e2313719121 (2024).

19. Yeo RW*, Zhou OY*, Zhong BL, **Sun ED**, Negredo PN, Nair S, Sharmin M, Ruetz TJ, Wilson M, Kundaje A, Dunn AR, Brunet A. Chromatin accessibility dynamics of neurogenic niche cells reveal defects in neural stem cell adhesion and migration during aging. **Nature Aging** 3, 866–893 (2023).

20. Ma R, **Sun ED**, Zou J. A spectral method for assessing and combining multiple data visualizations. **Nature Communications** 14, 780 (2023).

21. Chakrabarti A*, Michaels TCT*, Yin S, **Sun ED**, Mahadevan L. The cusp of an apple. **Nature Physics** 17, 1125–1129 (2021).

22. Lenormand T, Fyon F, **Sun ED**, Roze D. Sex chromosome degeneration by regulatory evolution. **Current Biology** 30, 3001-3006.e5 (2020).

Review Papers:

23. Ding A, **Sun ED**. Machine learning-based brain aging clocks: tools for deeper insight into neural aging and rejuvenation. In revision.

24. **Sun ED***, Nagvekar R*, Pogson AN*, Brunet A. Brain aging and rejuvenation at single-cell resolution. **Neuron** 113, 82–108 (2025).

Presentations

Seminars and talks:

– Invited speaker at Joint Statistical Meetings (JSM) session on "Frontiers in Statistics and Machine Learning for Decoding Aging Mechanisms". August 2026 (scheduled)

– Birnstiel laureate seminar at Institute for Molecular Pathology (IMP). November 2025.

– Invited presentation at NIA/NIH Longitudinal Studies Section meeting. September 2025.

- Invited presentation for mathematics topics group at Theis Lab. July 2025.
- Invited speaker at 2025 International Neural Regeneration Symposium Series Webinar. June 2025.
- Invited talk at STATGEN 2025 Session on Recent Developments in Computational Methods for Single-Cell Omics Integration and Modeling. May 2025.
- Invited panelist at ISTAART Basics Webinar. March 2025. "Immunity and Neurodegeneration: The Basics of Which Immune Cells are in the Brain and How do we Identify Them? A Single Cell Analysis Primer".
- Invited presentation for Statistics for Spatial Genomics Seminar. October 2024.
- Invited presentation at Gordian Biotechnology and Age1. October 2024.
- Invited seminar at Sunbelt Spatial Omics Seminar Series. October 2024.
- Proceedings talk at Intelligent Systems for Molecular Biology Conference. July 2024.
- Spotlight talk at Stanford AI for Health and Data Science Applications Workshop. April 2024.
- Research talk at Simons Collaboration on Plasticity and the Aging Brain Annual Meeting. October 2023.
- Research talk at Stanford Genetics Retreat. September 2023.
- Spotlight talk at ICML Computational Biology Workshop (**Top 10% of submissions**). July 2023.
- Research talk at Simons Collaboration on Plasticity and the Aging Brain Annual Meeting. April 2022.
- Research talk at SENS SRF Summer Symposium. September 2019.
- Oral presentation at American Physical Society March Meeting. March 2019.

Posters:

- Simons Collaboration on Plasticity and the Aging Brain Annual Meeting. October 2024.
- Cold Spring Harbor Laboratory Mechanisms of Aging. September 2024.
- Bay Area Aging Meeting (BAAM). May 2024. **Most Outstanding Poster Prize**
- Stanford DBDS Fest. May 2023.
- Stanford DBDS Fest. May 2022.
- Bay Area Aging Meeting (BAAM). May 2022.
- Stanford Biomedical Data Science Retreat. September 2021.
- Radcliffe Institute Research Day. April 2018. **Best Poster Prize**

Interviews:

- Invited interview. Aging clocks can predict how much older you are than your chronological age. How do they work?. LiveScience
- Invited interview. Eric Sun: Understanding brain aging at spatial and single-cell resolution with machine learning. Genomic Psychiatry
- Invited interview. The Single-cell and Spatial Playbook. Frontline Genomics

Teaching Experience

Workshop in Biostatistics (BIODS 260B): Guest Lecturer, Winter 2023
- Invited lecture titled "Meta-algorithms for high-dimensional data visualization"

Biomedical Data Science (BIOMEDIN 202): Teaching Assistant, Winter 2023
- Helped design new course, developed problem set for data fusion, lectured, held office hours, graded.

Data Science for Medicine (BIOMEDIN 215): Teaching Assistant, Fall 2021
- Assisted with in-class activities, held office hours, graded assignments.

Nonlinear Dynamical Systems (APMTH 108): Teaching Assistant, Spring 2019 & 2018
- Facilitated in-class activities, graded assignments, held office hours, taught exam review sessions.

Mentorship Experience

Massachusetts Institute of Technology
- Andrew Ding (MIT Biological Engineering PhD), Multi-organ single-cell aging clocks, Fall 2025-Present
- Yinuo Cheng (Harvard CompBio MS), Meta-learning & tabular foundation models, Fall 2025-Present
- David Goh (Harvard-MIT HST PhD Rotation), Spatial perturbation modeling, Winter 2025-Present
- Julia Holz (MIT CSB PhD Rotation), Spatial transcript spillover correction, Winter 2025-Present
- Amulya Garimella (MIT CSB PhD Rotation), Modeling cell replacement strategies, Winter 2025-Present
- Jessica Liu (Harvard Bioengineering PhD Rotation), Benchmarks for aging clocks, Winter 2025-Present
- Victoria Alkin (Harvard BBS PhD Rotation), Spatial domain search algorithms, Winter 2025-Present
- Steven Hui (UCSD undergraduate), AI agent knowledgebase for aging, Winter 2025-Present
- Kanon Kobata (MIT Microbiology PhD Rotation), MTA algorithms for yeast aging, Winter 2025
- Gaeun Kim (Harvard SSQB PhD Rotation), Spatially resolved single-cell foundation model, Fall 2025
- Sonny Young (MIT Chemistry PhD Rotation), Spatial domain search algorithms, Fall 2025

Stanford University
- Biomedical Data Science PhD Rotation, Spatial omics foundation model, Winter 2025
- Genetics PhD Rotation, Spatial profiling of corpora amylacea, Fall 2024
- Postbac Student, Pre-training for epigenetic aging clocks, 2024-2025

Harvard University
- Undergraduate, Quantifying heterogeneity in late-life mortality deceleration, 2019-2020

Service

Invited Peer Review: Bioinformatics, Computational and Structural Biotechnology Journal, Mechanisms of Ageing and Development, Nature, Nature Aging, Nature Communications, npj Aging, Science

Past & Current Society Memberships: American Aging Association (AGE), International Society for Computational Biology (ISCB), American Physical Society (APS)

Other:
- Proceedings Program Committee for ISMB 2026 conference

Outreach

Stanford Science Penpals: November 2022 - May 2023

ADVANCE Undergraduate Institute (AUI): April 2023

Pueblo Programming Workshops: 2015-2021